

Topics of today's class

Last class:

- Chapter 8
 - Income disparity among countries
 - Endogenous growth model

Today's class:

- Chapter 9
 - Consumer's consumption/savings decision
 - Responses of consumer to changes in income and interest rates.

Chapter 9

A Two-Period Model: The Consumption-Savings Decision and Government Deficits

Chapter 9 Topics

- Consumer's consumption/savings decision – responses of consumer to changes in income and interest rates.
- Government budget deficits and the Ricardian Equivalence Theorem.

Consumption

- $Y = C + I + G + NX$
- Consumption accounts for more than 2/3 of GDP.
- What key economic forces shape consumption/savings decisions?
- Will allow us to address several key issues:
 - Government Debts and Deficits
 - Financial Markets Imperfections / Financial Crisis
 - Social Security

Consumption

So far:

- Solow Model: $C = sY$
- Where does it come from? Why don't households optimize?
- With household optimization it becomes clear that:
 - expectations matters
 - the distinction between current and permanent income is crucial
 - real interest rate is a key variable

Dynamic Decision

- Previous micro-founded model of consumption and leisure is static.
- Life is full of inter-temporal choices:
 - Should I go to college or start working immediately?
 - Should I save or should I consume now?

The Model

Representative household lives 2 periods:

- “today” is period 1.
- “all future periods” are period 2.

Households receive exogenously

- income y in first period.
- income y' in second period (no labor supply choice).

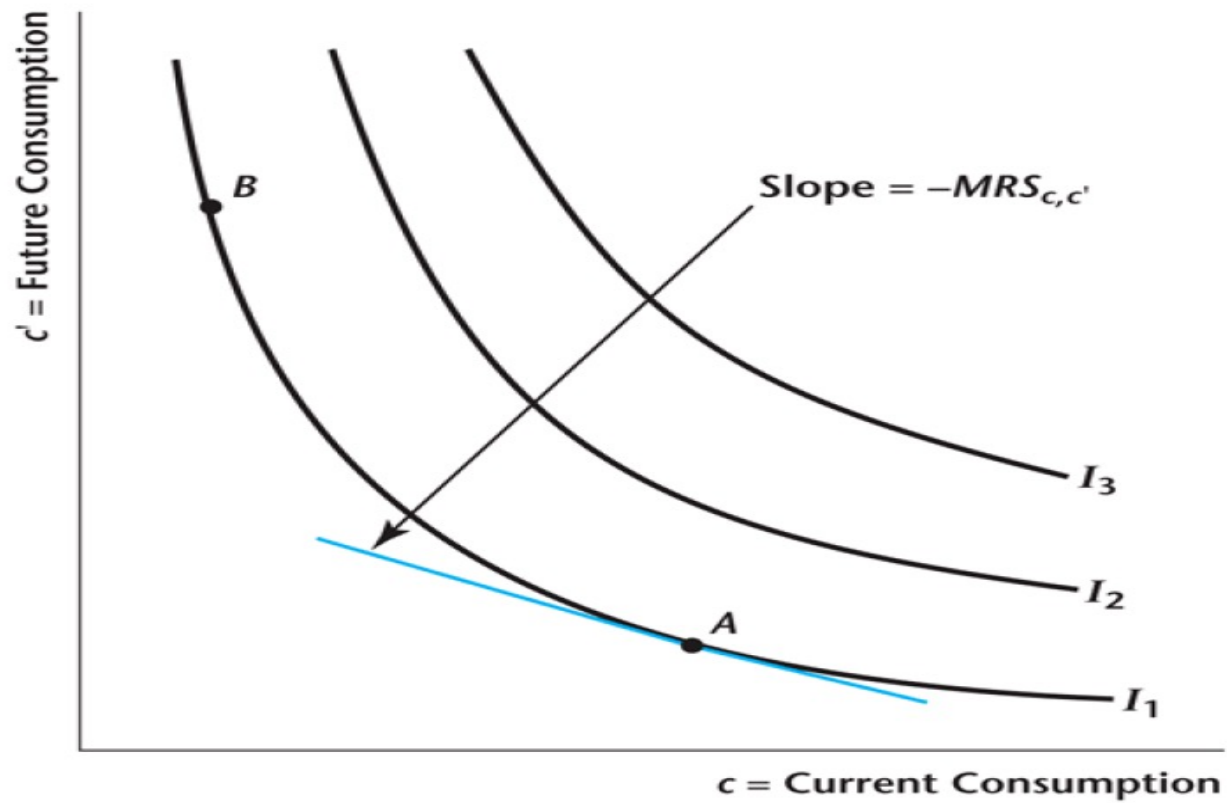
Utility Function $U(c, c')$

- c : first (current) period of life.
- c' : consumption in second (future) period of life.

Assumptions on preference

- Monotonicity: more is always preferred to less.
- Preference for Diversity also called Consumption smoothing.
- Current consumption and future consumption are normal goods.

Indifference curve



Inter-temporal Marginal Rate of Substitution

- The inter-temporal marginal rate of substitution is the rate at which the consumer is just willing to substitute current consumption for future consumption $MRS_{c,c'}$,
- Inter-temporal marginal rate of substitution measures willingness to substitute consumption over time.
- It decreases with current consumption.

Budget Constraints

The consumer's current-period budget constraint:

$$c + s = y - t$$

Budget Constraints

The consumer's future-period budget constraint:

$$c' = y' - t' + (1 + r)s$$

Consumer's Lifetime Budget Constraint

- Substitute in the current-period budget constraint obtaining lifetime budget constraint:

$$c + \frac{c'}{1+r} = y - t + \frac{y' - t'}{1+r}$$

- $\frac{1}{1+r}$: relative price of future consumption
 - If I give up 1 unit of consumption today, I can consume $1 + r$ unit of consumption tomorrow.
 - If I give up $\frac{1}{1+r}$ unit of consumption today, I can consume 1 unit of consumption tomorrow.

Consumer's Lifetime Wealth

$$we = y - t + \frac{y' - t'}{1 + r}$$

Simplified Lifetime Budget Constraint

$$c + \frac{c'}{1+r} = we$$

Simplified Lifetime Budget Constraint: Slope-Intercept

$$c' = -(1+r)c + we(1+r)$$

Figure 8.1

Consumer's Lifetime Budget Constraint

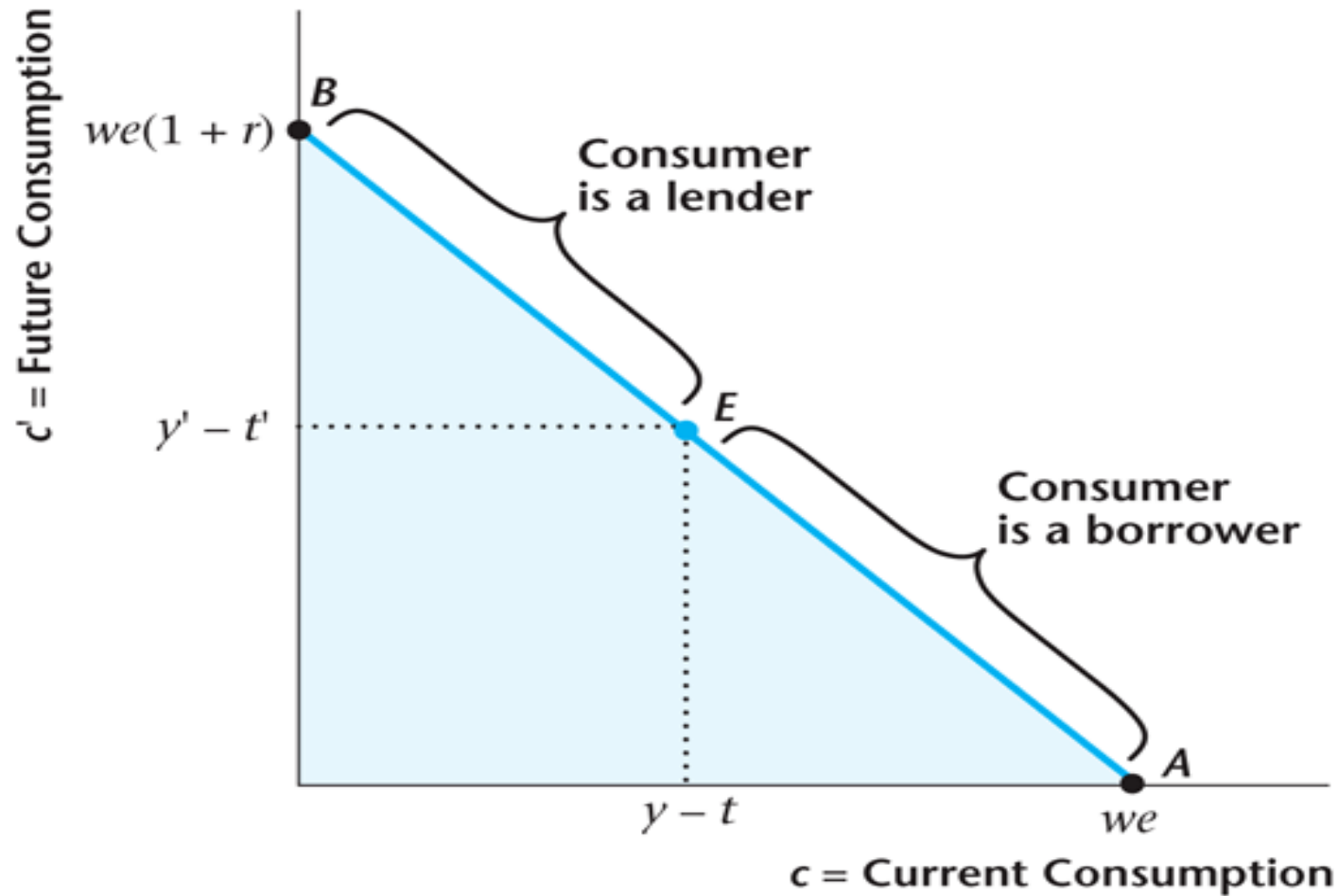


Figure 8.2

A Consumer's Indifference Curves

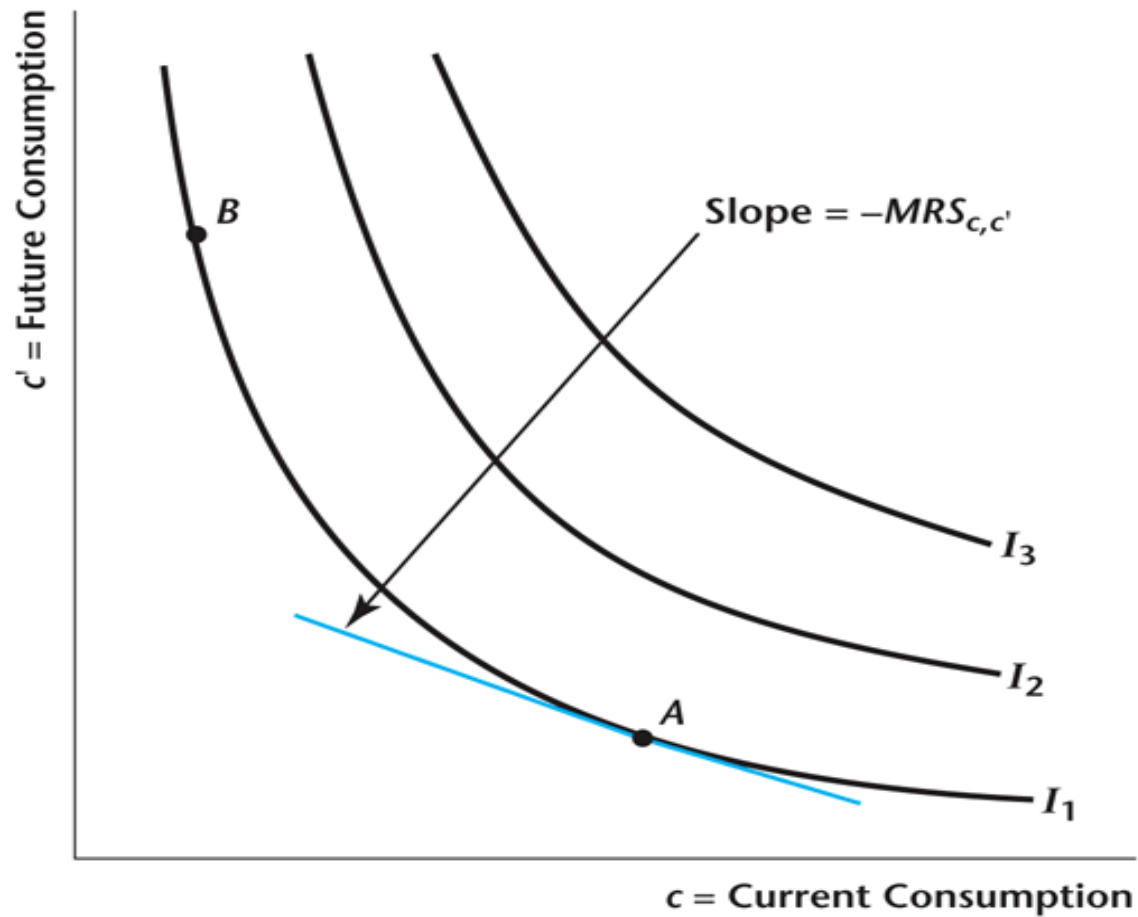


Table 8.1

Sara's Desire for Consumption Smoothing

Table 8.1 Sara's Desire for Consumption Smoothing

	Week 1 Coconuts	Week 2 Coconuts	Total Consumption
Bundle 1	5	15	20
Bundle 2	17	3	20
Preferred Bundle	11	9	20

Optimization

- Marginal condition that holds when the consumer is optimizing:

$$MRS_{c,c'} = 1 + r$$

- **Market:** 1 unit of c in exchange of $1 + r$ unit of c' .
- **Preferences:** 1 unit of c in exchange of $MRS_{c,c'}$ unit of c' .

Figure 8.3

A Consumer Who Is a Lender

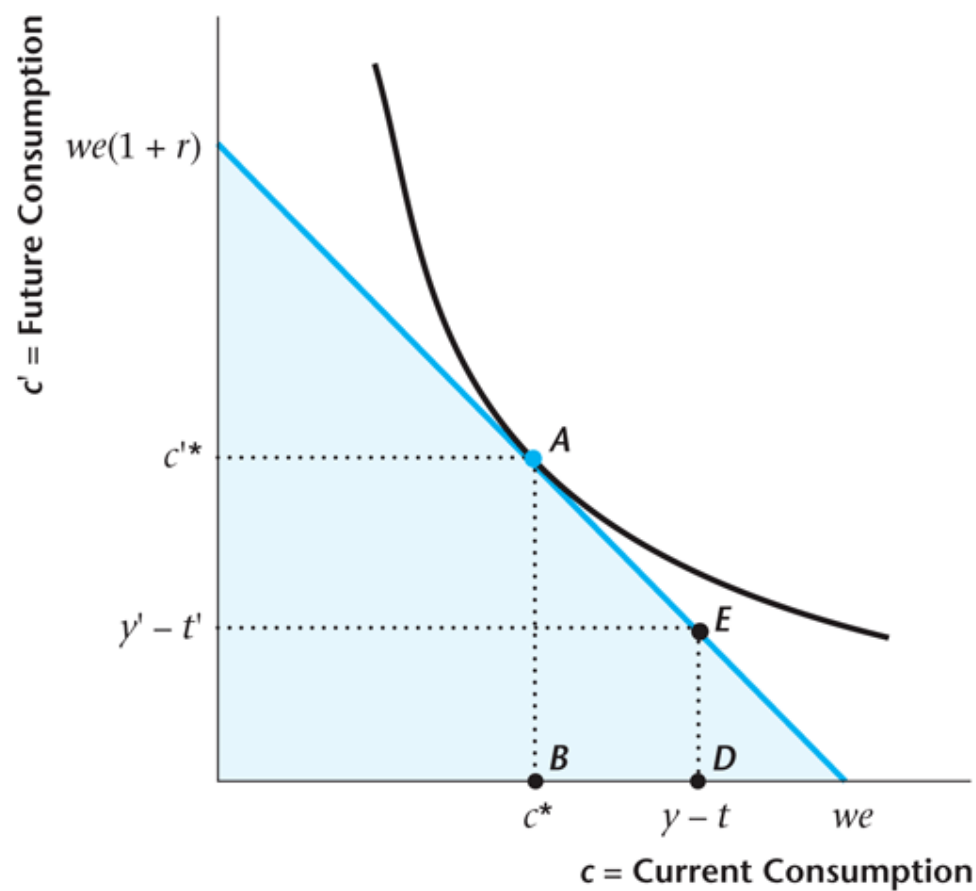
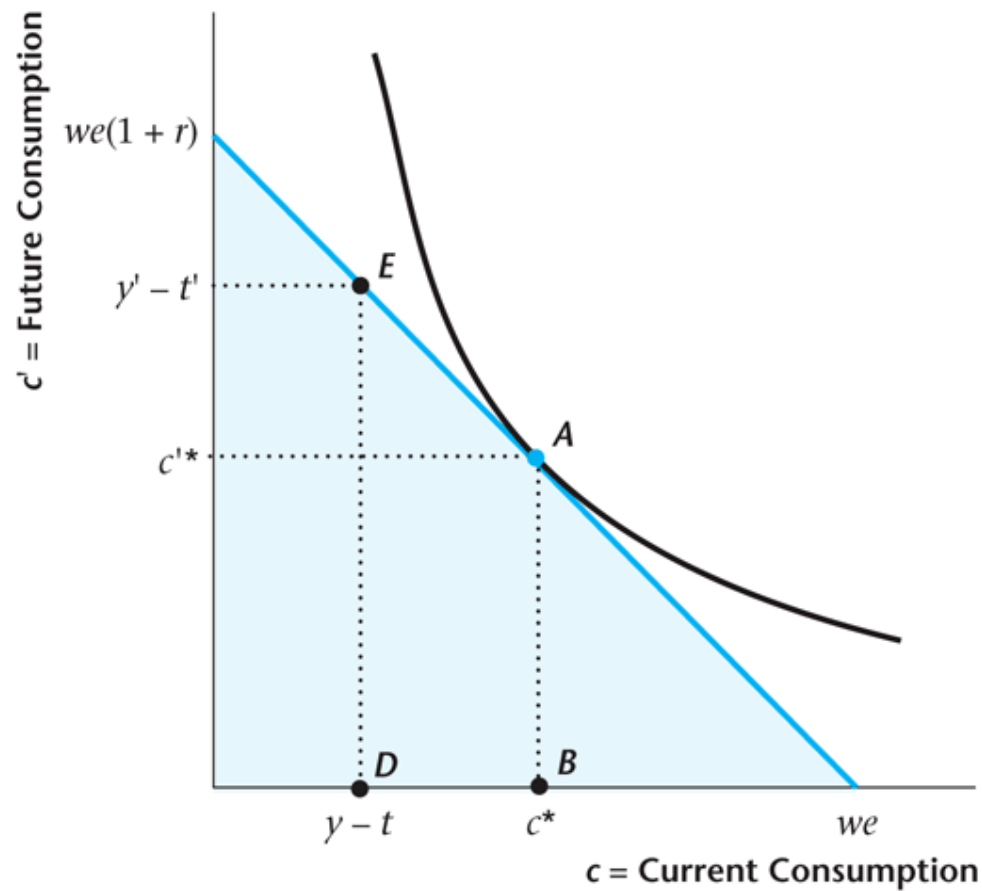


Figure 8.4

A Consumer Who Is a Borrower



Consumer Optimization

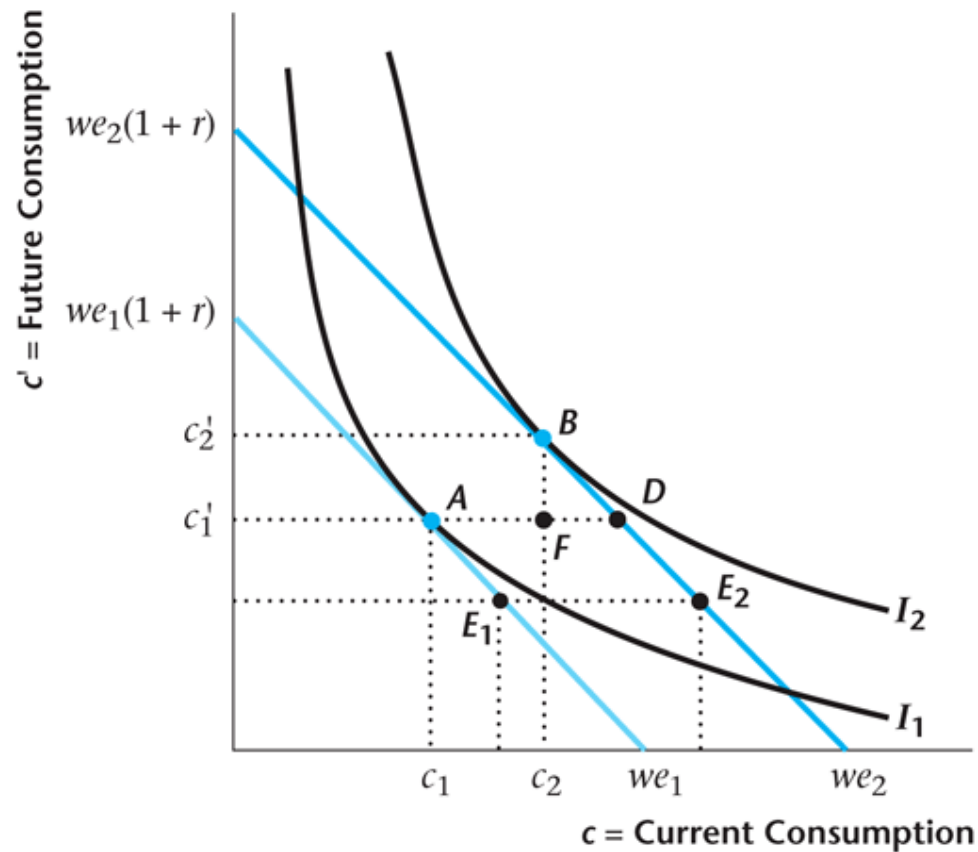
- Consumption Smoothing
 - The ability to borrow or lend creates a trade-off between current and future consumption.
 - c and c' depend only on lifetime wealth, w_e , not its composition.
- But saving *does depend* on composition of lifetime wealth:
 - Consumers with high first-period income are more likely to be savers (lenders).
 - Consumers with high second-period income are more likely to be borrowers, using future income to fund current consumption.
- Key assumption: Saving (borrowing) markets work perfectly.

An Increase in Current Period Income

- Example: Winning at the lottery. The consumer acts to smooth consumption over time:
 - Current and future consumption increase.
 - Saving increases because the increase in consumption is smaller than the increase in income.

Figure 8.5

The Effects of an Increase in Current Income for a Lender



Observed Consumption-Smoothing Behavior

- Aggregate consumption of non-durables and services is smooth relative to aggregate income, but the consumption of durables is more volatile than income.
- This is because durables consumption is economically more like investment than consumption.

Figure 8.6

Percentage Deviations from Trend in Consumption of Durables and Real GDP

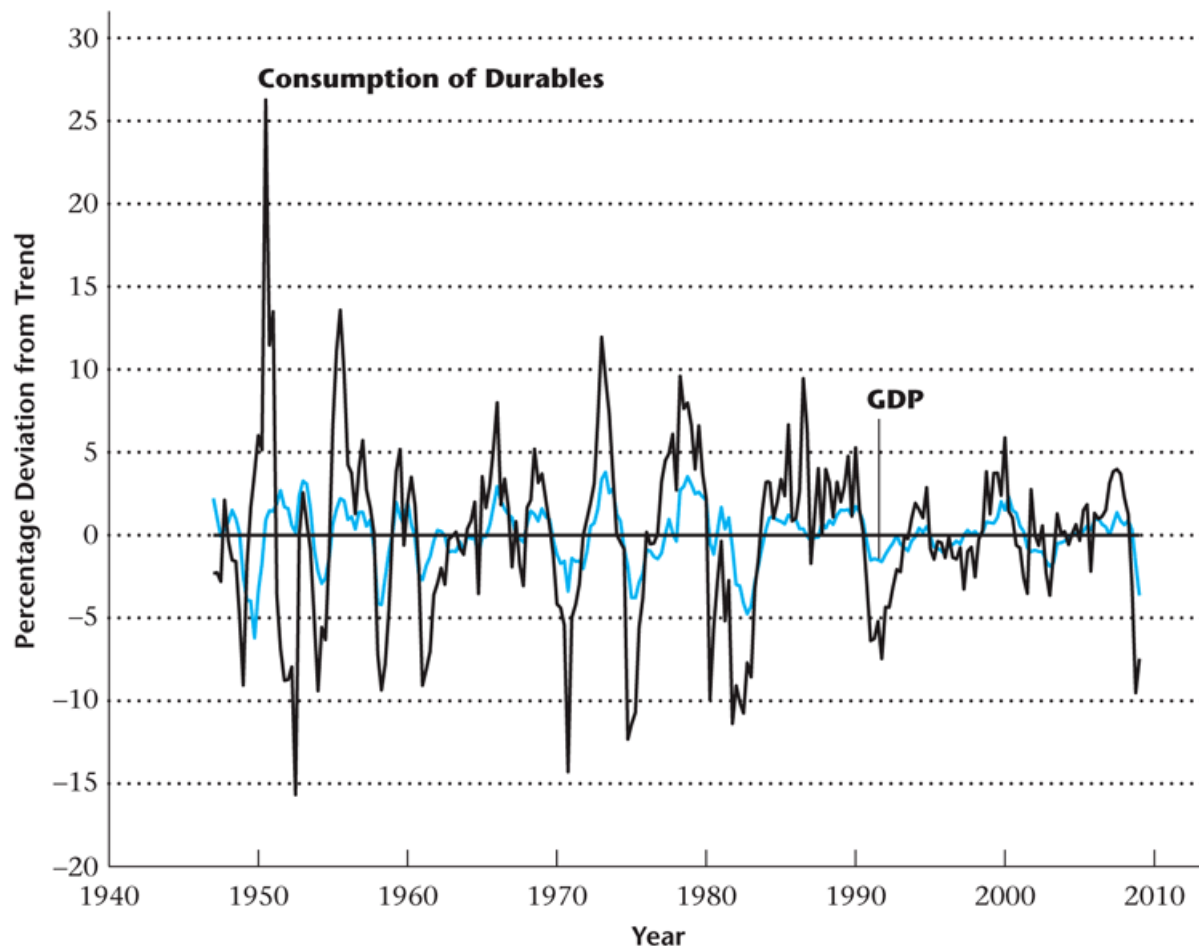
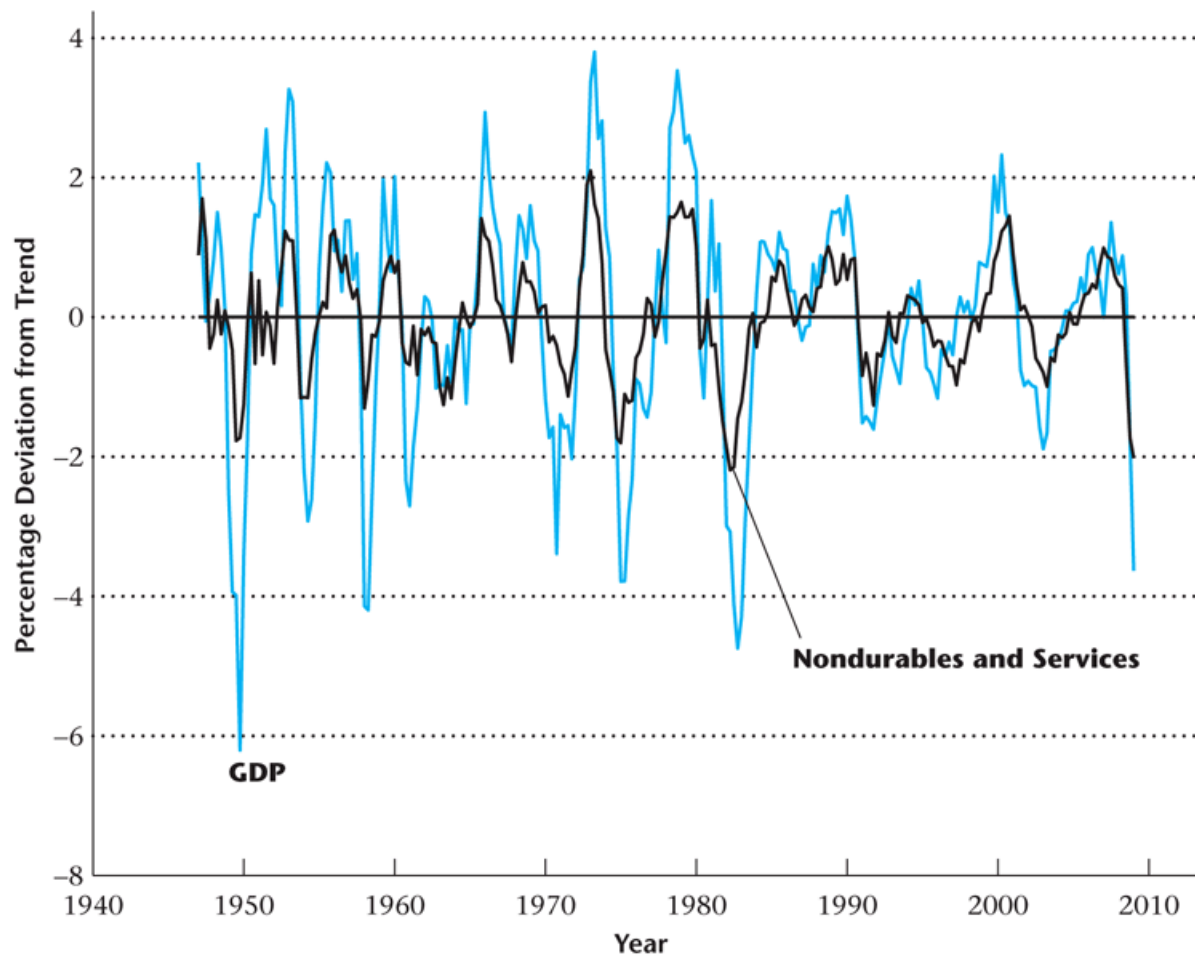


Figure 8.7

Percentage Deviations from Trend in Consumption of Nondurables and Services and Real GDP



An Increase in Future Income

Example: Changing Major: from philosophy to business

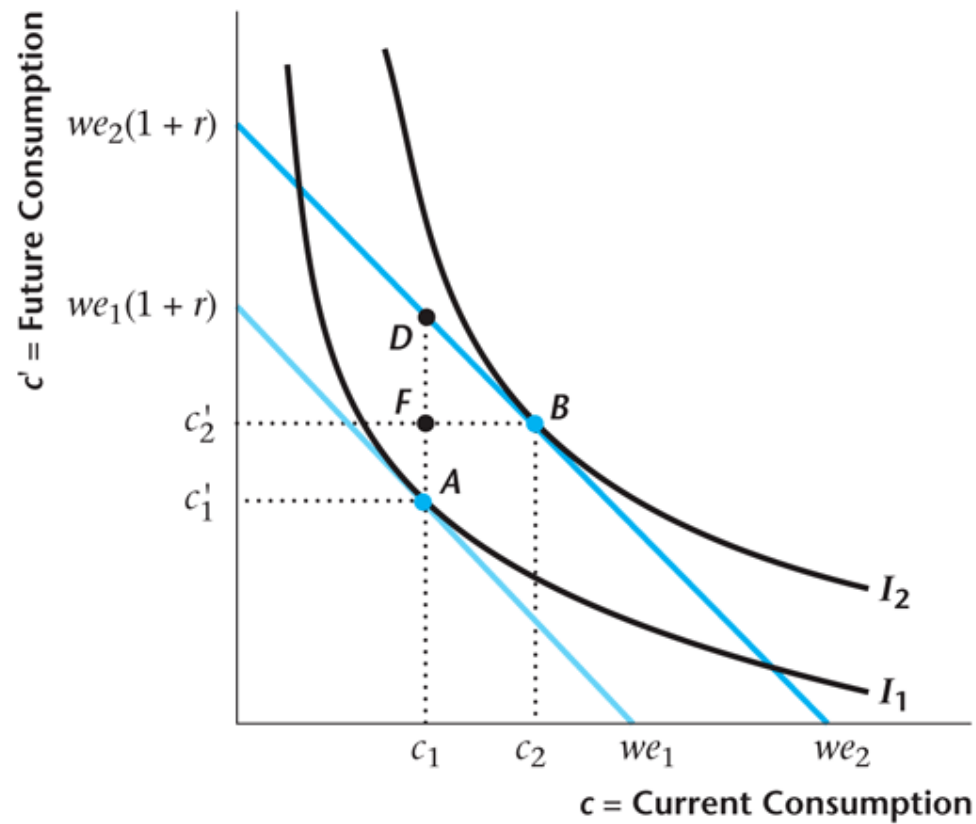
- c and c' : depend only on lifetime wealth, w_e , not its composition
- so, effects of an increase in y is the same as an increase of $y'/(1+r)$

Then,

- c and c' increase
- s decreases

Figure 8.8

An Increase in Future Income



Increase in the Real Interest Rate

1. Substitution Effects

2. Wealth Effects

Recall: $1/(1+r)$: relative price of c'

- decrease in the relative price of c' , consumption today become relatively more expensive
- higher return to saving induces more saving

Increase in the Real Interest Rate

1. Substitution Effects
2. **Wealth Effects**

Recall:

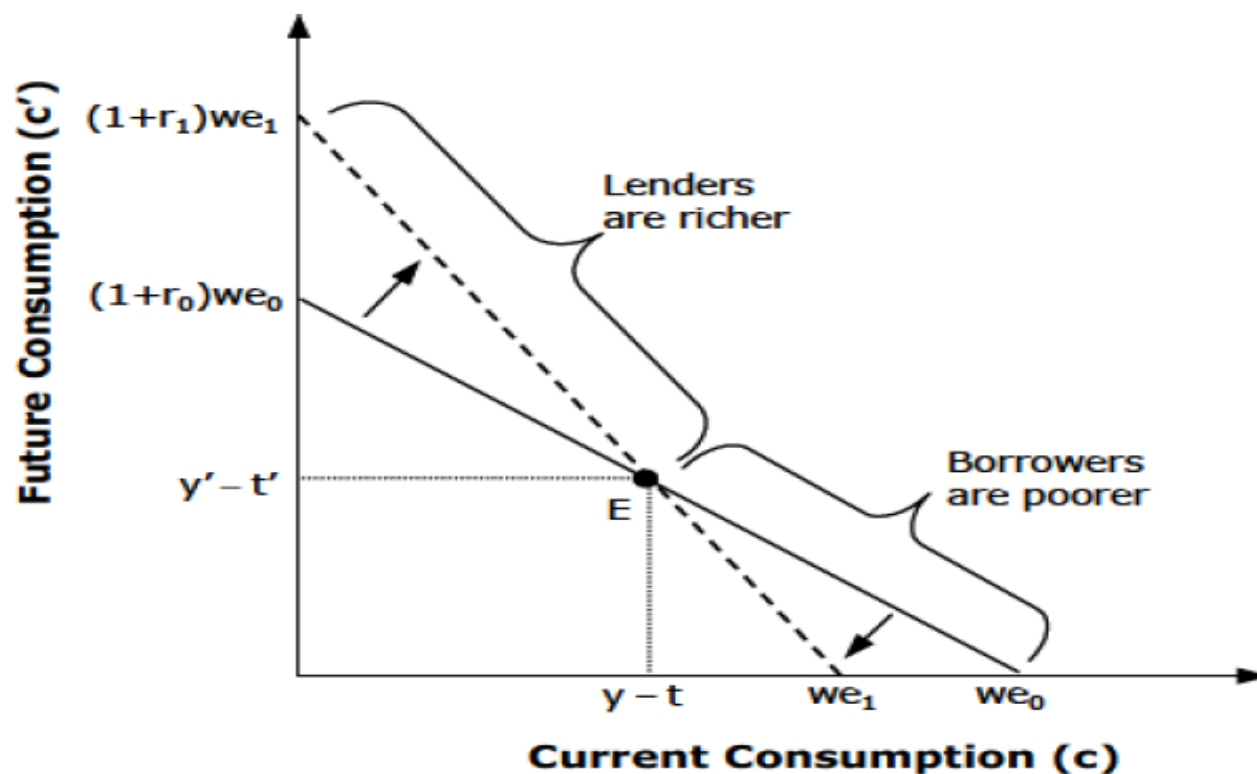
$$we = y - t + (y' - t')/(1 + r)$$
$$(1 + r)we = (1 + r)(y - t) + y' - t'$$

Increasing r :

- decreases we lifetime wealth
- increases $we(1 + r)$

Figure 8.12

An Increase in the Real Interest Rate

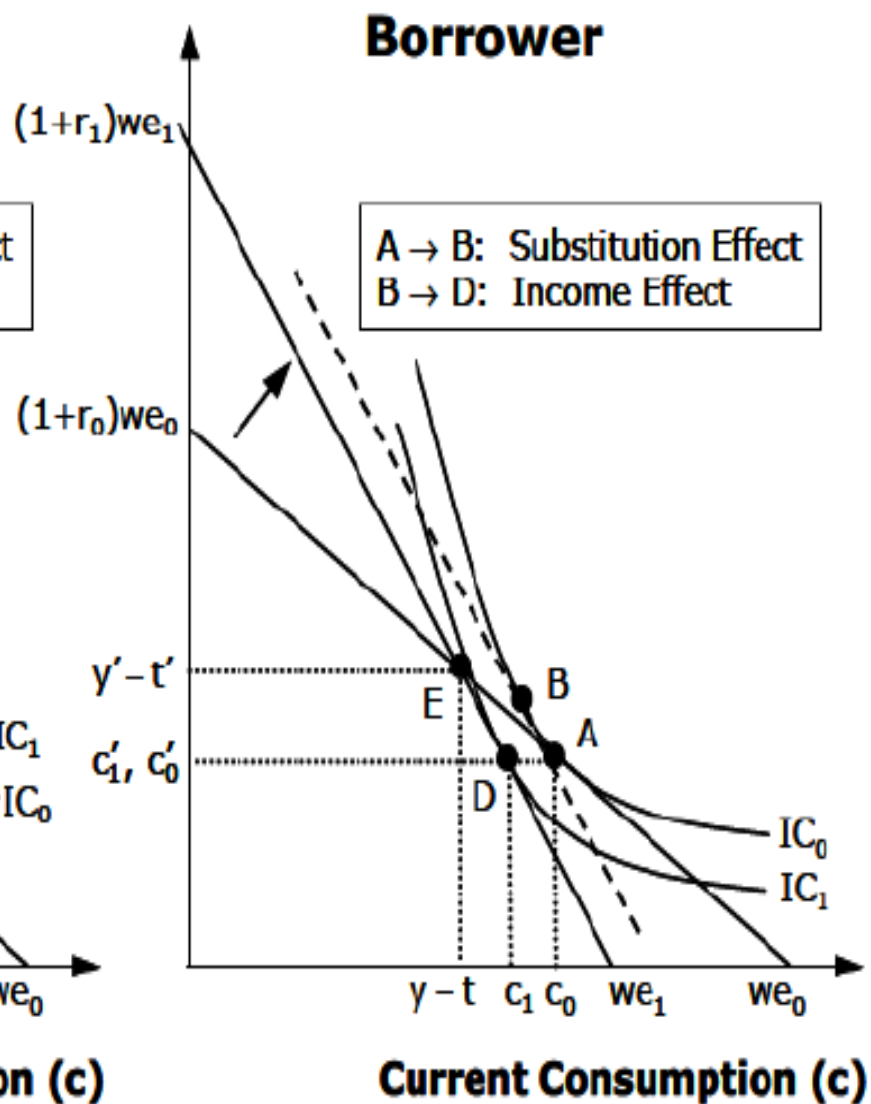
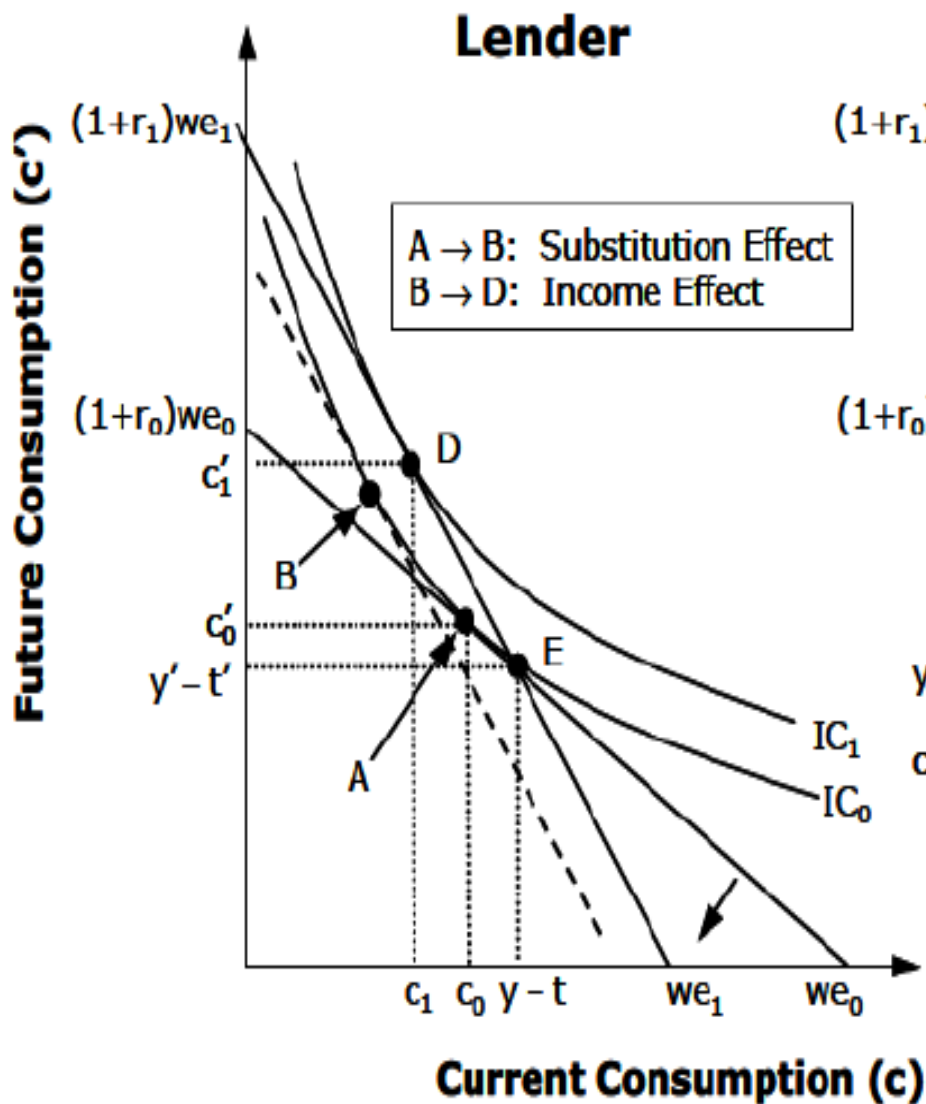


An Increase in the Market Real Interest Rate

1. Substitution Effects

2. Wealth Effects

- Lenders (savers): lenders are richer: both c and c' increase.
- Borrowers: c decreases and c' decreases



Effects of an Increase in the Real Interest Rate

Table 8.2 Effects of an Increase in the Real Interest Rate for a Lender

Current consumption	?
Future consumption	Increases
Current savings	?

Table 8.3 Effects of an Increase in the Real Interest Rate for a Borrower

Current consumption	Decreases
Future consumption	?
Current savings	Increases

Perfect Complements Example

- With perfect complements, the ratio of future consumption to current consumption is constant.

$$c' = ac$$

- The consumer's budget constraint must hold.

$$c + \frac{c'}{1+r} = we$$

Perfect Complements Example

- With perfect complements we can solve explicitly for current and future consumption:

$$c = \frac{we(1+r)}{1+r+a}$$

$$c' = \frac{awe(1+r)}{1+r+a}$$

Perfect Complements Example

- Substituting for lifetime wealth gives:

$$c = \frac{(y - t)(1 + r) + y' - t'}{1 + r + a}$$

$$c' = a \left[\frac{(y - t)(1 + r) + y' - t'}{1 + r + a} \right]$$

Figure 8.15

Example with Perfect Complements Preferences

